

Nonintegral Laplacian spectra of ideal intersection graphs

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Abstract

We study Laplacian integrality for the graph of nonzero proper ideals of $\mathbb{Z}_n$, where distinct ideals are adjacent when their intersection is nonzero. For squarefree composite n , the graph is Laplacian integral exactly when n has two prime factors. For three distinct primes we prove nonintegrality for every exponent vector (a, a, b) with $a, b \geq 1$, including $(1, 1, k)$, and for every three-prime exponent vector with minimum entry at most seven. For arbitrary ordered exponents $2 \leq a \leq b \leq c$, we also prove nonintegrality whenever $c \geq 4a^2 - 2a$. An inertia criterion also settles every triple whose maximum minus minimum is at most three and the family $(a, a+3, a+4)$ for all $a \geq 1$. A uniform complement eigenvalue bound in $(0, 3)$ further proves nonintegrality whenever $(a-2)(a+b+c-2) > 2bc$ and $a \geq 4$. In particular, every triple with minimum exponent at least three times its maximum-minus-minimum span plus eight is nonintegral. Arithmetic arguments also settle every triple with common exponent divisor at least three, every all-odd triple, and further congruence classes. Scaling and a divisibility contradiction also settle every triple whose three exponents are congruent to two modulo four, and hence every triple with a common 2-adic valuation. A further root-divisibility argument settles six mixed-parity exponent residue classes modulo eight and every permutation of $(3, 3, 2)$ modulo four, with no minimum or ratio bound. The integer endpoint two forces $c < 7a - 16 + 40/(a+2)$ for $a \geq 4$; all-even triples beyond this linear bound are therefore nonintegral. An exact endpoint certificate also settles every triple whose second-smallest exponent is at most fifteen. Endpoint reductions modulo three also settle residue triples $(2, 2, 2)$ and every permutation of $(1, 2, 2)$, with unbounded unequal exponents. The repeated-exponent proof combines arithmetic constraints on a quadratic discriminant with a uniform consecutive-integer interval for a symmetric cubic. Exact divisor reductions settle the few exceptional factor levels. Complement-quotient intervals prove the families with an exponent equal to one, two or three; the last case uses an exhaustive integer endpoint certificate after the uniform cutoff. An independent integer discriminant certificate exhausts the finite regions left by the cutoff for minimum entries four through seven. The inertia argument counts two low complement eigenvalues and excludes integer endpoints by four root-free modular certificates. The remaining unequal-exponent and nonsquarefree higher-prime cases stay open.

1 Introduction

The ideal intersection graph associates a graph to the intersection structure of the nonzero proper ideals of a ring [2]. For $\mathbb{Z}_n$, its isomorphism type depends on the exponents in the prime factorization of n . We ask which of these graphs have integer Laplacian eigenvalues.

Nikandish [1] develops a generalized-join formula for the Laplacian spectrum and a Schur–Weyl reduction for equal prime exponents. His results include nonintegrality for $(pqr)^k$ for every $k \geq 1$. His Question (Q3) asks for the full integrality characterization; the present arguments address this question for additional exponent families.

We prove a complete classification for squarefree composite integers (Theorem 3.1), an explicit quadratic obstruction for $(1, 1, k)$ (Theorem 4.1), and nonintegrality for every repeated-exponent

triple (a, a, b) , $a, b \geq 1$ (Theorem 7.1). The last result combines an arithmetic restriction on the antisymmetric block with a consecutive-integer sign interval for the symmetric cubic. The boundary family $b = (a-1)(2a-1)$ is proved in Theorem 5.1. An exact divisor criterion isolates every possible square discriminant for fixed a . A second divisor criterion fixes the smaller-factor index instead, giving a finite reduction over all a (Proposition 6.6). The first three nonboundary factor levels have complete finite modular certificates. Every later level lies strictly below the uniform sign threshold, closing the whole repeated-exponent family.

For a unit exponent, the complement of the general six-support quotient has a nonzero eigenvalue strictly between 0 and 1, except for three small pairs already covered by $(1, 1, k)$. This proves nonintegrality for every $(1, b, c)$ (Theorem 8.1), including fully unequal triples. Positive-coefficient expansions at a minimum exponent of two, together with one exceptional rational sign interval, then settle every $(2, b, c)$ (Theorem 9.1). Thus every three-prime exponent vector with minimum entry at most two is covered. The same endpoint polynomial gives a uniform cutoff $c \geq 4a^2 - 2a$ for ordered $2 \leq a \leq b \leq c$ (Proposition 10.1). For a minimum exponent of three, this leaves exactly 325 pairs. An exhaustive integer sign certificate, positive second-endpoint expansions and one rational exception settle all of them (Theorem 10.2). Thus every triple with minimum entry at most three is nonintegral.

For nearby unequal exponents, low roots can occur in pairs without an endpoint sign change. A symmetric Schur complement gives an inertia criterion for two roots in a specified interval (Proposition 11.1). At the upper endpoint three, four bounded-gap families satisfy the criterion for every $a \geq 4$. Positive first-endpoint factors and root-free second-endpoint factors exclude both possible integers. This proves Theorem 11.2, including the full family $(a, a+1, a+2)$ and every triple whose maximum minus minimum is at most three.

The requested diagnostic for minimum entries four through seven exhausts the finite region left by the uniform cutoff. All 27 562 quintics have positive nonsquare discriminants, verified by independent integer Sylvester determinants. Together with the written tail and repeated-exponent proofs, this yields Corollary 11.4: every triple with minimum entry at most seven is nonintegral. The dependence on the complete finite certificate is explicit.

The Schur complement also admits a uniform comparison that covers mixed diagonal signs. For every triple with minimum exponent at least four, the complement quotient has a positive eigenvalue below three (Theorem 12.2). If $(a-2)(a+b+c-2) > 2bc$, that eigenvalue lies in $(2, 3)$ (Corollary 12.3). This settles the growing balanced region where the minimum exponent is at least three times the exponent span plus eight (Corollary 12.4). The universal bound also reduces any possible integral triple in this range to the endpoint-zero condition $h_C(1)h_C(2) = 0$.

An integer-scaling argument then settles every triple with common exponent divisor at least three (Theorem 13.1). Finite-field factorization settles every all-odd triple and explicit common-residue classes, including residues 1, 3, 4 modulo five (Theorem 13.2). These arithmetic arguments allow unbounded exponent differences and do not require excluding all integer quotient roots. The distribution of three hypothetical odd quotient roots modulo four also forces divisibility by thirty-two at one or three, settling six mixed-parity exponent classes modulo eight (Theorem 13.6). A shifted-polynomial congruence then forces all three odd roots to be one modulo four for exponent residues $(3, 3, 2)$ modulo four. Incompatible endpoint and derivative divisibilities settle this entire family (Theorem 13.8).

The endpoint reductions modulo three further settle $(2, 2, 2)$ and all permutations of $(1, 2, 2)$ as exponent residue patterns (Theorem 14.5). Their endpoint values are both nonzero, so the positive root below three must be noninteger.

The graph and lifting arguments are given explicitly, so the proofs do not require a complete spectral decomposition. Section 16 describes the exact computations supporting the displayed

formulas and finite certificates. The classification for arbitrary exponent vectors remains open.

2 The graph and its invariant subspaces

Write $n = \prod_{i=1}^t p_i^{k_i}$, where the primes are distinct and $k_i \geq 1$. Denote by G_n the graph whose vertices are the nonzero proper ideals of $\mathbb{Z}_n$, with distinct ideals adjacent if their intersection is nonzero. We use the combinatorial Laplacian $L = D - A$; the graph is *Laplacian integral* when every eigenvalue of L is an integer, including when the graph is disconnected.

The ideal $(\prod_i p_i^{\alpha_i})$ is encoded by $0 \leq \alpha_i \leq k_i$. We exclude $(0, \dots, 0)$, which encodes the whole ring, and $(k_1, \dots, k_t)$, which encodes the zero ideal. Ideal intersection is generated by the least common multiple. Therefore two distinct vertices α, β are adjacent exactly when

$$\max(\alpha_i, \beta_i) < k_i \quad \text{for some } i.$$

Equivalently, putting $S(\alpha) = \{i : \alpha_i < k_i\}$, adjacency is $S(\alpha) \cap S(\beta) \neq \emptyset$.

For nonempty $S \subseteq [t]$, the support class $\mathcal{C}_S$ is a clique of size $\prod_{i \in S} k_i$, except that the full support has size $u = \prod_i k_i - 1$. Edges between distinct support classes are complete or absent according as their supports intersect or are disjoint. The full support consists of universal vertices. Remove them and call the resulting graph H ; thus $G_n = K_u \vee H$, with the evident interpretation for $u = 0$.

Lemma 2.1 (Lifting across the universal vertices). *If $L(H)v = \lambda v$ and $\lambda \neq 0$, extension of v by zero on the u universal vertices is an eigenvector of $L(G_n)$ with eigenvalue $\lambda + u$.*

Proof. Since $L(H)$ is symmetric with zero row sums, $\lambda \neq 0$ implies $\sum v = 0$. At a vertex of H , its degree increases by u and all new neighbor values are zero. At a universal vertex the Laplacian action is $-\sum v = 0$. These are exactly the claimed eigenvector equations. $\square$

For $(k_1, k_2, k_3) = (a, a, b)$, H has six support classes of sizes a, a, b, a^2, ab, ab on $1, 2, 3, 12, 13, 23$, respectively. The partition into

$$X = \mathcal{C}_1 \cup \mathcal{C}_2, \quad Y = \mathcal{C}_3, \quad Z = \mathcal{C}_{12}, \quad W = \mathcal{C}_{13} \cup \mathcal{C}_{23}$$

is equitable, and on functions constant on these cells the Laplacian acts by

$$B = \begin{pmatrix} a^2 + ab & 0 & -a^2 & -ab \\ 0 & 2ab & 0 & -2ab \\ -2a & 0 & 2a + 2ab & -2ab \\ -a & -b & -a^2 & a^2 + a + b \end{pmatrix}. \quad (1)$$

For example, an X vertex has a^2 neighbors in Z and ab in W , while a W vertex has a, b, a^2 neighbors in X, Y, Z . Internal neighbors cancel for cell-constant functions. The cell sizes are $2a, b, a^2, 2ab$, and multiplying B on the left by their diagonal matrix gives a symmetric matrix. Thus B is similar to a real symmetric matrix.

The functions taking values $s, -s$ on $\mathcal{C}_1, \mathcal{C}_2$, $t, -t$ on $\mathcal{C}_{13}, \mathcal{C}_{23}$, and zero elsewhere form a second invariant subspace, with matrix

$$B_- = \begin{pmatrix} a^2 + ab & -ab \\ -a & a^2 + a + b + 2ab \end{pmatrix}. \quad (2)$$

These functions have sum zero. Their eigenvalues therefore also lift by $u = a^2b - 1$, directly by the proof of Lemma 2.1.

Expanding (1) gives $\det(xI - B) = xf_{a,b}(x)$, where

$$\begin{aligned} f_{a,b}(x) &= x^3 - E_1x^2 + E_2x - E_3, \\ E_1 &= 2a^2 + 5ab + 3a + b, \\ E_2 &= a^4 + 7a^3b + 3a^3 + 8a^2b^2 + 11a^2b + 2a^2 + 3ab^2 + 2ab, \\ E_3 &= 2a^2b(a + b + 1)(a^2 + 2ab + 2a + b). \end{aligned} \tag{3}$$

Its roots are real and nonzero: reality follows from the similarity above, and $E_3 > 0$. Every root λ gives an eigenvalue $\lambda + u$ of G_n . A noninteger root of $f_{a,b}$ therefore proves nonintegrality.

3 Squarefree integers

Theorem 3.1. *For squarefree composite n , the graph G_n is Laplacian integral if and only if n has exactly two prime factors.*

Proof. Let t be the number of prime factors. For $t = 2$ the graph consists of the two ideals (p) and (q) , with no edge, so $L = 0$. Assume $t \geq 3$. The vertices are the nonempty proper subsets of $[t]$, adjacent when they intersect. On the tensor space encoding all subsets put

$$E = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}, \quad F = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \quad J = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad c = 2^t - 1.$$

The principal submatrix of $\Phi = cI - E^{\otimes t} + F^{\otimes t} - J^{\otimes t}$ on the graph vertices is $L(G_n)$: the diagonal entry at S is $2^t - 2 - 2^{t-|S|}$, and the off-diagonal entry is -1 for intersecting subsets and zero otherwise.

With e_0, e_1 denoting the absence and presence basis, set $w = e_0 \otimes e_1 - e_1 \otimes e_0$. For every 2×2 matrix T , one has $T^{\otimes 2}w = (\det T)w$. A paired factor w thus contributes $2, -1, 0$ under E, F, J , respectively. It vanishes at the empty and full subsets. Each subspace used below is invariant for Φ and is supported on graph vertices, so its restriction is also a restriction of L .

If $t = 2r + 1$, $r \geq 1$, use $w^{\otimes r} \otimes \text{span}\{e_0, e_1\}$. Putting $a = 2^r$ and $s = (-1)^r$, the restricted matrix is

$$\begin{pmatrix} c - 2a + s & s \\ s & c - a \end{pmatrix},$$

with discriminant $(a - s)^2 + 4$. The integer $h = a - s$ is at least 3, so $h^2 < h^2 + 4 < (h + 1)^2$. Both eigenvalues are irrational.

If $t = 2r$, $r \geq 2$, use $r - 1$ paired factors and the symmetric subspace on the last two coordinates, with basis $e_0e_0, (e_0e_1 + e_1e_0)/\sqrt{2}, e_1e_1$. Put $a = 2^{r-1}$ and $s = (-1)^{r-1}$. The restriction is $cI - K$, where

$$K = \begin{pmatrix} 4a - s & -s\sqrt{2} & -s \\ -s\sqrt{2} & 2a - s & 0 \\ -s & 0 & a \end{pmatrix}.$$

The leading principal minors of $K - (a - 1)I$ are

$$3a + 1 - s, \quad (3a + 1 - s)(a + 1 - s) - 2, \quad (3a - s)(a + 1 - s) - 2.$$

They are positive for $a \geq 2$ and $s = \pm 1$: the latter two are at least 10 and 8, respectively. Sylvester's criterion gives $\lambda_{\min}(K) > a - 1$. The principal submatrix of $K - aI$ on coordinates 1, 3 is $\begin{pmatrix} 3a - s & -s \\ -s & 0 \end{pmatrix}$, with determinant -1 . A vector supported on those coordinates has a negative Rayleigh quotient, giving $\lambda_{\min}(K) < a$. Hence $c - \lambda_{\min}(K)$ is an eigenvalue of L strictly between the consecutive integers $c - a$ and $c - a + 1$. $\square$

4 The family pqr^k

Theorem 4.1. *For distinct primes p, q, r and every $k \geq 1$, the graph G_{pqr^k} has the irrational eigenvalues*

$$\frac{6k+1 \pm \sqrt{(2k+2)^2 - 3}}{2}.$$

In particular, every permutation of the exponent vector $(1, 1, k)$ gives a nonintegral Laplacian spectrum.

Proof. The support classes 1, 2 have size 1, and 13, 23 have size k . Consider functions with values $s, -s$ on the first two classes, $t, -t$ on the latter two, and zero elsewhere. A vertex of support 1 has $2k$ neighbors and Laplacian value $2ks - kt$. A vertex of support 13 has degree $4k$; after subtracting the same-class and opposite-class values its Laplacian value is $-s + (4k+1)t$. The swapped classes give their negatives, and all other classes give zero by cancellation. The restricted matrix of the full graph is therefore

$$B_k = \begin{pmatrix} 2k & -k \\ -1 & 4k+1 \end{pmatrix}.$$

Its characteristic polynomial is $x^2 - (6k+1)x + 8k^2 + k$, with discriminant $(2k+2)^2 - 3$. Since

$$(2k+1)^2 < (2k+2)^2 - 3 < (2k+2)^2$$

for $k \geq 1$, the two roots are irrational. Relabeling prime coordinates preserves the graph. $\square$

5 A boundary family with integral antisymmetric block

The following argument was supplied by Reza Nikandish in the public collaboration discussion. We include it because it is the boundary case needed for the linear cutoff in Section 6.

Theorem 5.1. *For every $a \geq 2$, put $C = (a-1)(2a-1)$. For distinct primes p, q, r , the graph $G_{p^a q^a r^C}$ is not Laplacian integral, although the antisymmetric block (2) has integer eigenvalues.*

Proof. Specialize $f_{a,b}$ from (3) to $b = C$. For $a = 2$ it is $x^3 - 47x^2 + 702x - 3312$. Its values at 0, 1, 2, 3, 4 modulo 5 are 3, 4, 2, 3, 3. Thus it has no root in $\mathbb{F}_5$ and is irreducible over $\mathbb{Q}$; its real roots are noninteger. Lemma 2.1 gives noninteger graph eigenvalues.

For $a \geq 3$ put $m = 2a^3 - 2a^2 + 2a - 1$. Expansion gives

$$\begin{aligned} f_{a,C}(m) &= 2(5a^4 - 10a^3 + 7a^2 - 1), \\ f_{a,C}(m-1) &= -2(a-1)(a+2)(2a^4 - 7a^3 + 7a^2 - a - 3). \end{aligned}$$

The first value is positive because its parenthesis is $5a^3(a-2) + 7a^2 - 1$. Writing $z = a-3 \geq 0$, the last factor in the second value is $2z^4 + 17z^3 + 52z^2 + 68z + 30 > 0$. Consequently $f_{a,C}(m-1) < 0 < f_{a,C}(m)$, so a real root lies strictly between $m-1$ and m . Its integer shift by $u = a^2C - 1$ is noninteger.

The characteristic discriminant of B_- is $D = (a+1)^2b^2 + 2a(3a+1)b + a^2$. At $b = C$ it equals $(2a^3 - a^2 + a - 1)^2$, and the two eigenvalues are $4a^3 - 3a^2 + a$ and $2a^3 - 2a^2 + 1$. Thus the obstruction in this family comes from the symmetric cubic. $\square$

6 Finite reduction and a linear cutoff

Fix $a, b \geq 1$, and put

$$h = a + 1, \quad C = (a - 1)(2a - 1), \quad M = a^3(2a + 1), \quad A = h^2b + a(3a + 1).$$

The characteristic discriminant of (2) is

$$D = h^2b^2 + 2a(3a + 1)b + a^2. \quad (4)$$

If D is not a square, this block has irrational eigenvalues, which remain irrational after the integer shift $a^2b - 1$.

Proposition 6.1 (Exact divisor reduction). *For fixed $a \geq 1$, the positive integers b for which D is a square are exactly those obtained from positive factor pairs $d \leq e$ of M such that*

$$b = \frac{d + e - a(3a + 1)}{h^2} \in \mathbb{Z}_{\geq 1}, \quad s = \frac{e - d}{h} \in \mathbb{Z}_{\geq 0}. \quad (5)$$

In that case $D = s^2$.

Proof. The polynomial identity

$$A^2 - h^2D = 4M \quad (6)$$

holds for arbitrary a, b . If $D = s^2$ with $s \geq 0$, then $A - hs$ and $A + hs$ are positive: their product is positive and their sum is $2A > 0$. They have the same parity; since their product is divisible by 4, both are even. Write $A - hs = 2d$ and $A + hs = 2e$. Then $de = M$, $d + e = A$ and $e - d = hs$, giving (5). Conversely those conditions yield $A = d + e$ and $hs = e - d$. Using $(d + e)^2 - (e - d)^2 = 4de = 4M$ in (6) gives $D = s^2$. $\square$

The divisor set is finite for every fixed a ; this criterion has no truncation in b . It gives first a quadratic bound.

Corollary 6.2. *If D is a square then $b \leq C$. Consequently every $b > C$ gives a nonintegral Laplacian spectrum. For $a = 1$, every $b \geq 1$ is covered.*

Proof. Since $(d - 1)(e - 1) \geq 0$, $A = d + e \leq M + 1$. The identity

$$M + 1 - a(3a + 1) = h^2C$$

then gives $b \leq C$. The nonsquare case is covered by the antisymmetric block. At $a = 1$, $C = 0$. $\square$

Lemma 6.3. *Every admissible factor pair in Proposition 6.1 satisfies $d \equiv e \equiv 1 \pmod{h}$.*

Proof. From $e - d = hs$ and $d + e = A$, reduction modulo h gives $e \equiv d$ and $2d \equiv 2$, since $a(3a + 1) \equiv 2 \pmod{h}$. Also

$$(d - 1)(e - 1) = M - A + 1 = h^2(C - b). \quad (7)$$

If h is odd, the congruence $2(d - 1) \equiv 0$ gives the assertion. If h is even, the alternative to $d \equiv 1$ is $d - 1 \equiv h/2 \pmod{h}$. Then $d - 1$ and $e - 1$ would both equal $h/2$ times an odd integer. Their product divided by h^2 would be an odd integer divided by 4, contradicting (7). This excludes the alternative for either parity of a . $\square$

Theorem 6.4 (Unified linear cutoff). *Let $a \geq 2$ and*

$$R(a) = \frac{2(a-1)(a-2)}{a+2} = 2a - 10 + \frac{24}{a+2}.$$

For every integer $b > R(a)$ and distinct primes p, q, r , the graph $G_{p^a q^a r^b}$ is not Laplacian integral. Every unresolved repeated-exponent pair therefore has square D and $1 \leq b \leq \lfloor R(a) \rfloor$.

Proof. If D is nonsquare the antisymmetric block suffices. Otherwise use Proposition 6.1. When $d = 1$, $e = M$ and $b = C$; Theorem 5.1 settles this case. If $d > 1$, Lemma 6.3 gives $d \geq T = a + 2$. Since $e \geq d \geq T$,

$$\frac{M}{T} + T - (d + e) = (d - T) \left(\frac{e}{T} - 1 \right) \geq 0.$$

Hence

$$b \leq \frac{M/T + T - a(3a + 1)}{h^2} = \frac{2(a-1)(a-2)}{a+2} = R(a).$$

Thus $b > R(a)$ leaves either a nonsquare discriminant or the already settled boundary case. $\square$

At $(a, b) = (4, 2)$ the square-discriminant pair has $d = 6 = a + 2$, so equality in the factor estimate is attained. The graph is still nonintegral, as shown in the next section.

6.1 Settling the first nonboundary factor

The first possible value after $d = 1$ is $d = a + 2$. Since d divides M , the identity

$$M = (a + 2)(2a^3 - 3a^2 + 6a - 12) + 24$$

forces $a + 2 \mid 24$. With $a \geq 2$ this leaves $a = 2, 4, 6, 10, 22$. Substitution into (5) gives $b = 0, 2, 5, 12, 35$, respectively. Excluding $b = 0$, the complete positive- b list and its cubic certificates are:

(a, b)	$f_{a,b}(x)$	Root-free prime
$(4, 2)$	$x^3 - 86x^2 + 2304x - 18816$	5
$(6, 5)$	$x^3 - 245x^2 + 19266x - 488160$	7
$(10, 12)$	$x^3 - 842x^2 + 230160x - 20534400$	13
$(22, 35)$	$x^3 - 4919x^2 + 7887858x - 4132479120$	17

Each polynomial has no root modulo its indicated prime. Its monic cubic is therefore irreducible over $\mathbb{Q}$, and its real nonzero roots lift to noninteger graph eigenvalues. Thus every possible $d = a + 2$ pair is settled, over all a , by this finite divisor argument.

Theorem 6.5 (Second linear cutoff). *For $a \geq 2$ put*

$$S(a) = \frac{(a-3)(2a-3)}{2a+3} = a - 6 + \frac{27}{2a+3}.$$

For every positive integer $b > S(a)$ and distinct primes p, q, r , $G_{p^a q^a r^b}$ is not Laplacian integral. Every remaining square-discriminant case has $d \geq 2a + 3$ and $b \leq \lfloor S(a) \rfloor$.

Proof. Nonsquare D is covered by the antisymmetric block. If D is square, the cases $d = 1$ and $d = a + 2$ are already settled. Every other admissible factor satisfies $d \geq T = 2a + 3$ by Lemma 6.3. Since $e \geq d \geq T$, the same factor inequality gives

$$b \leq \frac{M/T + T - a(3a + 1)}{(a + 1)^2} = \frac{(a-3)(2a-3)}{2a+3} = S(a).$$

This proves the claim for $b > S(a)$. $\square$

The new bound is strictly smaller than $R(a)$ for all $a \geq 2$, since

$$R(a) - S(a) = \frac{2a^3 - a^2 - a - 6}{(a+2)(2a+3)} > 0.$$

Writing $z = a - 2 \geq 0$, its numerator is $2z^3 + 11z^2 + 19z + 4$.

6.2 A finite reduction for each factor index

For a nonboundary square-discriminant pair, write $d = 1 + hj$, $j \geq 1$. The following criterion fixes j instead of a .

Proposition 6.6. *Fix $j \geq 1$ and set $K_j = (j+1)^3(j+2)$. Every admissible square-discriminant pair of index j is obtained from a positive divisor d of K_j with $a = (d-1)/j - 1 \in \mathbb{Z}_{\geq 2}$. Set $e = M/d$, $v = (e-1)/(a+1)$, and retain exactly those candidates with $d \leq e$ and $b = C - jv \geq 1$. Then e, v, b are integers and $D = (v-j)^2$. Conversely every retained candidate is admissible.*

Proof. Modulo $d = j(a+1) + 1$ we have $ja \equiv -(j+1)$, whence

$$j^4 M = (ja)^3 j(2a+1) \equiv (j+1)^3(j+2) = K_j \pmod{d}.$$

Since $\gcd(d, j) = 1$, we have $d \mid M$ if and only if $d \mid K_j$. Thus e is integral. Since $M \equiv d \equiv 1 \pmod{h}$, also $e \equiv 1 \pmod{h}$, so v is integral. The identity

$$M = 1 + h(3a-2) + h^2 C = (1+hj)(1+hv)$$

gives $j + v + hjv = 3a - 2 + hC$. Defining $b = C - jv$, we obtain $d + e = h^2 b + a(3a+1)$ and $e - d = h(v-j)$. The ordering condition gives $v-j \geq 0$; Proposition 6.1 now proves admissibility and $D = (v-j)^2$. The reverse direction follows by applying these identities to any admissible index- j pair. $\square$

This is a complete finite problem for each fixed j , without a truncation in a . Section 7 combines the first three such problems with a uniform argument for all later indices. For $j = 2$, $K_2 = 108$ and $d = 2a + 3$ is an odd divisor of 108 with $d \geq 7$. The only candidates are $d = 9, 27$, giving $a = 3, 12$. The first has $e = 21, v = 5, b = 0$ and is excluded. The second has $e = 1600, v = 123, b = 7$ and $\sqrt{D} = 121$. Thus $(12, 7)$ is the only positive- b case with $j = 2$.

Its symmetric cubic is $f_{12,7}(x) = x^3 - 751x^2 + 180348x - 13829760$. Its values at all residues modulo 13 are 4, 7, 9, 3, 8, 4, 10, 6, 11, 5, 7, 10, 7. As a monic cubic with no root modulo 13, it is irreducible over $\mathbb{Q}$. Its real nonzero roots lift to noninteger graph eigenvalues, settling the entire index-2 case.

Theorem 6.7 (Third linear cutoff). *For every $a \geq 2$, put $T(a) = 2(a-4)(a-2)/(3a+4)$. If $b \geq 1$ and $b > T(a)$, then $G_{p^a q^a r^b}$ is not Laplacian integral. Every unresolved square-discriminant pair has $j \geq 3$, $d \geq 3a+4$, and $b \leq \lfloor T(a) \rfloor$.*

Proof. Nonsquare D gives irrational antisymmetric eigenvalues. For square D , the cases $j = 0, 1, 2$ are all settled above. Every other pair has $d \geq Q = 3a+4$, and $e \geq d \geq Q$ gives

$$b \leq \frac{M/Q + Q - a(3a+1)}{(a+1)^2} = \frac{2(a-4)(a-2)}{3a+4} = T(a).$$

This proves the claimed implication. $\square$

For $a \geq 3$ the new bound is strictly below $S(a)$, since

$$S(a) - T(a) = \frac{2a^3 - a^2 - 5a - 12}{(2a + 3)(3a + 4)} > 0.$$

With $z = a - 3 \geq 0$, the numerator is $2z^3 + 17z^2 + 43z + 18$. The exact expression $T(a) = 2a/3 - 44/9 + 320/(9(3a + 4))$ shows its leading term.

7 Completion of the repeated-exponent family

Theorem 7.1. *For distinct primes p, q, r and all integers $a, b \geq 1$, the graph $G_{p^a q^a r^b}$ is not Laplacian integral. The same conclusion holds for every permutation of the exponent vector (a, a, b) .*

The arithmetic reduction and a uniform sign obstruction for the cubic combine to exhaust the family. We first settle one last finite factor level.

Lemma 7.2. *Every square-discriminant pair with index $j = 3$ gives a nonintegral Laplacian spectrum.*

Proof. Proposition 6.6 gives $d \mid K_3 = 320$ and $d = 3a + 4$. For $a \geq 2$, the complete divisor list of this shape is 10, 16, 40, 64, 160. The resulting values are

d	a	e	v	b	Outcome
10	2	4	1	0	$d > e$
16	4	36	7	0	$b = 0$
40	12	1080	83	4	Root-free modulo 5
64	20	5125	244	9	Root-free modulo 7
160	52	92274	1741	30	Root-free modulo 7

The three positive- b cubics are

$$\begin{aligned} f_{12,4}(x) &= x^3 - 568x^2 + 100032x - 5248512, \\ f_{20,9}(x) &= x^3 - 1769x^2 + 992820x - 174744000, \\ f_{52,30}(x) &= x^3 - 13394x^2 + 57771168x - 80229951360. \end{aligned}$$

Their values at all residues of the indicated primes are, respectively, $(3, 3, 3, 4, 2)$, $(3, 2, 4, 1, 6, 4, 1)$ and $(1, 6, 4, 1, 3, 2, 4)$. Each monic cubic is therefore irreducible over $\mathbb{Q}$ and has noninteger real roots. Section 2 supplies the lift to noninteger graph eigenvalues. $\square$

Lemma 7.3 (A uniform consecutive-integer interval). *For $a, b \geq 1$, put*

$$L(a) = \frac{a^3 + 2a^2 + 2a + 1}{a(2a + 1)}.$$

If $b < L(a)$, then $f_{a,b}$ has a root strictly between $2ab - 1$ and $2ab$.

Proof. Substitution in (3) gives

$$\begin{aligned} f_{a,b}(2ab) &= 2a^2b^2 > 0, \\ f_{a,b}(2ab - 1) &= -(a + b + 1)(a^3 + 2a^2 + 2a + 1 - a(2a + 1)b). \end{aligned}$$

The second value is strictly negative when $b < L(a)$. The intermediate value theorem gives the asserted noninteger, nonzero root. $\square$

Proof of Theorem 7.1. For $a = 1$, Corollary 6.2 settles every positive b . For $a \geq 2$, nonsquare D gives irrational antisymmetric eigenvalues. When D is square, index $j = 0$ is Theorem 5.1, and indices $j = 1, 2$ are settled in Section 6. Lemma 7.2 settles $j = 3$.

It remains to consider $j \geq 4$. Then $d \geq Q = 4a + 5$, and $e \geq d \geq Q$ implies

$$\frac{M}{Q} + Q - (d + e) = (d - Q) \left(\frac{e}{Q} - 1 \right) \geq 0.$$

Since $d + e = (a + 1)^2 b + a(3a + 1)$, we obtain

$$b \leq B_4(a) = \frac{M/Q + Q - a(3a + 1)}{(a + 1)^2} = \frac{(a - 5)(2a - 5)}{4a + 5}.$$

For every $a \geq 2$,

$$L(a) - B_4(a) = \frac{41a^3 - 17a^2 - 11a + 5}{a(2a + 1)(4a + 5)} > 0.$$

Indeed, with $z = a - 2 \geq 0$, the numerator becomes $41z^3 + 229z^2 + 413z + 243$. Thus $b \leq B_4(a) < L(a)$, and Lemma 7.3 gives a noninteger cubic root, which lifts across the universal vertices. These cases exhaust all positive a, b . Permuting the prime factors preserves the graph's isomorphism type. $\square$

The preceding cutoffs and fixed-exponent theorems are intermediate consequences of the arithmetic reduction. The uniform interval now settles the infinite range left by those cutoffs, without a parameter scan.

8 A unit exponent with two arbitrary exponents

Theorem 8.1. *For distinct primes p, q, r and every $b, c \geq 1$, the graph $G_{pq^b r^c}$ is not Laplacian integral. The same holds for permutations of the exponent vector $(1, b, c)$.*

The equal-exponent decomposition is replaced by a complement identity on all six support classes. For arbitrary $a, b, c \geq 1$, remove the $u = abc - 1$ universal vertices and denote the remaining graph by H . In the order $1, 2, 3, 12, 13, 23$, its class-size vector is $w = (a, b, c, ab, ac, bc)^T$ and its vertex count is $N = a + b + c + ab + ac + bc$. Disjoint supports are adjacent in the complement $\overline{H}$, whose cell-constant Laplacian quotient is

$$C = \begin{pmatrix} b + c + bc & -b & -c & 0 & 0 & -bc \\ -a & a + c + ac & -c & 0 & -ac & 0 \\ -a & -b & a + b + ab & -ab & 0 & 0 \\ 0 & 0 & -c & c & 0 & 0 \\ 0 & -b & 0 & 0 & b & 0 \\ -a & 0 & 0 & 0 & 0 & a \end{pmatrix}.$$

Multiplication on the left by $\text{diag}(w)$ makes C symmetric. Moreover $C\mathbf{1} = 0$ and $w^T C = 0$. If B is the quotient of H , the complement relation gives

$$B + C = NI - \mathbf{1}w^T. \tag{8}$$

Thus an eigenvector $Cv = \mu v$ with $\mu \neq 0$ satisfies $w^T v = 0$ and $Bv = (N - \mu)v$. When $N - \mu \neq 0$, Lemma 2.1 lifts it to an eigenvalue $N + u - \mu$ of G_n .

Proof of Theorem 8.1. Write $\det(xI - C) = xh_{a,b,c}(x)$. Direct determinant expansion yields

$$\begin{aligned} h_{a,b,c}(0) &= -abc(a + b + c)N, \\ h_{a,b,c}(a) &= -a^2b^2c^2(a^2 + 2a - b - c). \end{aligned}$$

At $a = 1$, the first value is strictly negative and

$$h_{1,b,c}(1) = b^2c^2(b + c - 3).$$

When $b + c > 3$, a real root μ lies strictly between 0 and 1. Equation (8) and the universal-vertex lift produce a Laplacian eigenvalue $V - \mu \in (V - 1, V)$, where $V = N + u$ is the integer vertex count. This eigenvalue is noninteger. When $b + c \leq 3$, the positive pairs are $(1, 1), (1, 2), (2, 1)$; Theorem 4.1 settles the corresponding permutations of $(1, 1, k)$. Permuting the prime factors preserves the graph's isomorphism type. $\square$

The general six-support reduction also gives an exact diagnostic for fully distinct triples. For $1 \leq a < b < c \leq 7$, all thirty-five quintics have a nonlinear irreducible factor over $\mathbb{Q}$. The quintics at $(1, 2, 7)$ and $(2, 3, 5)$ have factor degrees $(1, 4)$, and that at $(2, 4, 6)$ has factor degrees $(2, 3)$; irreducibility of the entire quintic is therefore not a uniform obstruction. Theorem 8.1 covers every triple beginning with 1 without a range restriction; the thirty-five-case calculation itself remains finite evidence.

9 A minimum exponent of two

Theorem 9.1. *For distinct primes p, q, r and all $b, c \geq 1$, the graph $G_{p^2q^br^c}$ is not Laplacian integral. Consequently every three-prime exponent vector with minimum entry at most two gives a nonintegral graph.*

Proof. Use the complement quotient and quintic from Section 8. For $a = 2$, $h_{2,b,c}(0) < 0$. Putting $s = b + c$ and $t = bc$ gives

$$h_{2,b,c}(1) = (s - 13)t^2 - (2s + 6)t + 3s^3 + s^2 - 3s - 1.$$

By symmetry order $b \leq c$. A unit exponent is covered by Theorem 8.1; if $b = 2$ or $b = c$, use Theorem 7.1. It remains to consider $3 \leq b < c$.

When $b = 3$ and $c = 5 + v$, $v \geq 0$, the exact expansion is

$$h_{2,3,5+v}(1) = 12v^3 + 112v^2 + 268v + 120 > 0.$$

When $b \geq 4$ and $c > b$, write $b = 4 + u$ and $c = 5 + u + v$, with $u, v \geq 0$. Then

$$\begin{aligned} h_{2,4+u,5+u+v}(1) &= (u^2 + 8u + 19)v^3 \\ &\quad + (4u^3 + 38u^2 + 128u + 170)v^2 \\ &\quad + (5u^4 + 62u^3 + 271u^2 + 502u + 368)v \\ &\quad + 2u^5 + 32u^4 + 190u^3 + 504u^2 + 552u + 160 > 0. \end{aligned}$$

All coefficients are positive. In either region the opposite signs at 0 and 1 give a root $\mu \in (0, 1)$ and, by (8) and the universal-vertex lift, a noninteger graph eigenvalue in $(V - 1, V)$.

The only remaining pair is $(b, c) = (3, 4)$. Here

$$h_{2,3,4}(x) = x^5 - 53x^4 + 953x^3 - 6523x^2 + 13134x - 7560,$$

and $h_{2,3,4}(1) = -48$ while $h_{2,3,4}(3/2) = 13437/32 > 0$. A root lies in $(1, 3/2)$. Since $V = 58$, its lifted graph eigenvalue lies in $(113/2, 57)$, an interval containing no integer. This exhausts all positive b, c . Combining with Theorem 8.1 proves the minimum-entry assertion. $\square$

10 A uniform cutoff and a minimum exponent of three

Proposition 10.1. *For distinct primes p, q, r and positive exponents ordered as $2 \leq a \leq b \leq c$, the graph $G_{p^a q^b r^c}$ is not Laplacian integral whenever $c \geq 4a^2 - 2a$.*

Proof. Use the complement quintic $h = h_{a,b,c}$ of Section 8. Its first endpoint is the cubic

$$h(1) = Lc^3 + Qc^2 + Rc + S, \quad (9)$$

where

$$\begin{aligned} L &= a^2 + b^2 - 1, \\ Q &= a^3 - 4a^2b^2 + 2a^2b - a^2 + 2ab^2 + ab - 3a + b^3 - b^2 - 3b + 3, \\ R &= (a-1)(b-1)(2ab + 3a + 3b - 3), \\ S &= (a-1)(b-1)(a+b-1)(ab + a + b - 1). \end{aligned}$$

Here $L, R, S > 0$. In fact the sharper condition $Lc + Q \geq 0$ already gives $h(1) > 0$. Put $T(a) = 4a^2 - 2a$. Then

$$Q + T(a)L = b^2(b-1) + (2a^2 + a - 3)b + K(a),$$

where $K(a) = 4a^4 - a^3 - 5a^2 - a + 3$. For $a = 2 + u$, $u \geq 0$,

$$K(2+u) = 4u^4 + 31u^3 + 85u^2 + 95u + 37 > 0.$$

Consequently $Lc + Q > 0$ for $c \geq T(a)$. Since $h(0) < 0$, a root lies in $(0, 1)$, producing a noninteger graph eigenvalue in $(V-1, V)$ by (8) and Lemma 2.1. $\square$

Theorem 10.2. *For distinct primes p, q, r and all $b, c \geq 1$, the graph $G_{p^3 q^b r^c}$ is not Laplacian integral. Consequently every three-prime exponent vector with minimum entry at most three gives a nonintegral graph.*

Proof. Theorems 8.1, 9.1 and 7.1 leave $4 \leq b < c$. Proposition 10.1 settles $c \geq 30$. At $a = 3$, equation (9) is

$$\begin{aligned} h_{3,b,c}(1) &= (b^2 + 8)c^3 + (b-1)(b^2 - 30b - 12)c^2 \\ &\quad + 6(b-1)(3b+2)c + 4(b-1)(b+2)(2b+1). \end{aligned}$$

The remaining domain $4 \leq b < c < 30$ has exactly $\binom{26}{2} = 325$ pairs. Exhaustive integer evaluation of this cubic gives 257 positive and 68 negative values, with no zero. Table 1 lists every negative range; all other pairs in this finite domain have positive values. The complete polynomial and separate Horner sign certificates are in `results/distinct-tail.json`; their construction is described in Section 16. Thus a positive value gives a root in $(0, 1)$.

For the negative values, the second endpoint is positive except at $(b, c) = (4, 5)$. Indeed, for $v \geq 0$,

$$h_{3,4,6+v}(2) = 104v^3 + 1238v^2 + 4018v + 2344 > 0.$$

For $b \geq 5$, write $b = 5 + u$, $c = 6 + u + v$, with $u, v \geq 0$. Direct expansion gives

$$\begin{aligned} h_{3,5+u,6+u+v}(2) &= (5u^2 + 54u + 153)v^3 \\ &\quad + (20u^3 + 262u^2 + 1179u + 1863)v^2 \\ &\quad + (25u^4 + 426u^3 + 2642u^2 + 7011u + 6624)v \\ &\quad + 10u^5 + 218u^4 + 1828u^3 + 7200u^2 + 12600u + 6480 > 0. \end{aligned}$$

b	4	5	6	7	8	9	10	11	12	13
$c_{\min}$	5	6	7	8	9	10	11	12	13	14
$c_{\max}$	13	15	16	17	17	16	16	15	14	14

Table 1: Every negative $h_{3,b,c}(1)$ in $4 \leq b < c < 30$; each column includes all integers between its two endpoints.

These regions exhaust $4 \leq b < c$ except $(4, 5)$. Whenever $h(1) < 0 < h(2)$, a root lies in $(1, 2)$, giving a noninteger graph eigenvalue in $(V - 2, V - 1)$.

Finally

$$h_{3,4,5}(x) = x^5 - 83x^4 + 2327x^3 - 24133x^2 + 60576x - 42480.$$

Here $h(1) = -3792$ and $h(3/2) = 48825/32 > 0$. The root in $(1, 3/2)$ lifts to a graph eigenvalue in $(233/2, 117)$, since $V = 118$. Every interval used contains no integer. The finite nonvanishing certificate and the written infinite-range arguments exhaust all b, c ; Theorems 8.1 and 9.1 give the final assertion. $\square$

11 An inertia criterion and four families with bounded exponent gaps

An endpoint sign change need not detect two roots in the same interval. For example, the complement quintic at equal exponents factors as

$$h_{a,a,a}(x) = (x - a^2 - a)(x^2 - (a^2 + 4a)x + 3a^2)^2.$$

We instead count low eigenvalues by a symmetric Schur complement.

Proposition 11.1. *Order the exponents as $a \leq b \leq c$, and let m be an integer with $2 \leq m < a$. If*

$$(c - m)(a + b + c - m) < mab, \quad (10)$$

then the six-support complement quotient has exactly two nonzero eigenvalues in $(0, m)$. If, in addition,

$$(a - m + 1)(a + b + c - m + 1) > (m - 1)bc, \quad (11)$$

both lie in $(m - 1, m)$, so $G_{p^a q^b r^c}$ is not Laplacian integral.

Proof. Put $s = a + b + c$ and $P = abc$. Symmetrize C by the square roots of its cell sizes and reorder the classes as 1, 2, 3, 23, 13, 12. The resulting real symmetric matrix is

$$C = \begin{pmatrix} A & -\sqrt{P}I \\ -\sqrt{P}I & D \end{pmatrix}, \quad D = \text{diag}(a, b, c),$$

where A has diagonal $(b + c + bc, a + c + ac, a + b + ab)$ and off-diagonal entries $A_{ij} = -\sqrt{a_i a_j}$, with $(a_1, a_2, a_3) = (a, b, c)$. For $0 < x < a$, the block $D - xI$ is positive definite, and Sylvester's law of inertia gives

$$\text{inertia}(C - xI) = \text{inertia}(D - xI) + \text{inertia}(K(x)),$$

where

$$K(x) = \text{diag}(k_a(x), k_b(x), k_c(x)) - vv^T, \quad v = (\sqrt{a}, \sqrt{b}, \sqrt{c})^T, \quad k_t(x) = s - x - \frac{xP}{t(t - x)}.$$

For fixed $x > 0$, the function $k_t(x)$ increases with $t > x$, since

$$\frac{\partial k_t(x)}{\partial t} = \frac{xP(2t-x)}{t^2(t-x)^2} > 0.$$

Condition (10) is exactly $k_c(m) < 0$. Thus all three diagonal values are negative, so $K(m)$ is negative definite. The matrix $C - mI$ has three negative and three positive eigenvalues. The complement support graph is a triangle with one pendant cell at each vertex; it is connected, so C has a simple zero eigenvalue and all other eigenvalues are positive. Exactly two nonzero eigenvalues therefore lie in $(0, m)$.

For $x = m - 1$, condition (11) says $k_a(x) > 0$, so all three $k_t(x)$ are positive. Congruence by the inverse square roots of these values turns $K(x)$ into $I - zz^T$, where

$$\|z\|^2 = \frac{a}{k_a(x)} + \frac{b}{k_b(x)} + \frac{c}{k_c(x)} > 1,$$

because each $k_t(x) < s$. Consequently $K(x)$ has one negative and two positive eigenvalues, and the only eigenvalue of C below x is zero. Both low positive eigenvalues lie in $(m - 1, m)$. They are less than $a < N$, so the complement identity and universal-vertex lift apply. Their graph eigenvalues lie in $(V - m, V - m + 1)$, which contains no integer. $\square$

Theorem 11.2. *For every $a \geq 1$, each of the exponent triples*

$$(a, a + 1, a + 2), \quad (a, a + 1, a + 3), \quad (a, a + 2, a + 3), \quad (a, a + 3, a + 4)$$

gives a nonintegral ideal intersection graph on three distinct primes.

Proof. For $a \leq 3$, use Theorem 10.2. Assume $a \geq 4$ and write the triple as $(a, a + d, a + e)$, where $(d, e) = (1, 2), (1, 3), (2, 3), (3, 4)$. The left side minus the right side of (10), at $m = 3$, is respectively

$$-6a, \quad -2a, \quad -4a, \quad 4 - 2a.$$

These values are negative. Proposition 11.1 therefore places exactly two nonzero complement eigenvalues in $(0, 3)$. It remains to exclude the integers 1 and 2.

The first endpoints have the identities $h_{a,a+d,a+e}(1) = -4M_{d,e}(a)P_{d,e}(a)$ from Table 2. Every multiplier is positive for $a \geq 4$, and every coefficient of $P_{d,e}(4 + u)$ is positive. Thus $h(1) < 0$ in each family.

(d, e)	$M_{d,e}(a)$	Descending coefficients of $P_{d,e}(4 + u)$
$(1, 2)$	$a(a + 1)$	$(1, 18, 116, 314, 294)$
$(1, 3)$	$a(a + 1)(a + 3)$	$(1, 13, 51, 57)$
$(2, 3)$	$a + 2$	$(1, 25, 242, 1126, 2484, 2022)$
$(3, 4)$	$a + 3$	$(1, 28, 302, 1550, 3699, 3126)$

Table 2: Exact first-endpoint identities and positive shifted factors.

The second endpoints factor as

$$\begin{aligned} h_{a,a+1,a+2}(2) &= -aF_{1,2}(a), \\ h_{a,a+1,a+3}(2) &= -(a - 2)(a + 1)F_{1,3}(a), \\ h_{a,a+2,a+3}(2) &= -aF_{2,3}(a), \\ h_{a,a+3,a+4}(2) &= -F_{3,4}(a), \end{aligned}$$

where

$$\begin{aligned}
F_{1,2}(a) &= a^5 - 9a^4 - 14a^3 + 93a^2 + 67a + 6, \\
F_{1,3}(a) &= a^4 - 6a^3 - 42a^2 - 39a - 10, \\
F_{2,3}(a) &= a^5 - 5a^4 - 48a^3 - 59a^2 - 39a - 54, \\
F_{3,4}(a) &= a^6 - a^5 - 74a^4 - 375a^3 - 967a^2 - 1220a - 340.
\end{aligned}$$

Table 3 evaluates each factor at every residue of one prime. No value is zero, so no factor has an integer root. The linear multipliers are nonzero for $a \geq 4$, hence $h(2) \neq 0$. Both positive eigenvalues in $(0, 3)$ are therefore noninteger, and their graph lifts are noninteger as well.

Factor	Prime	Values at residues $0, \dots, p-1$
$F_{1,2}$	7	(6, 4, 1, 5, 6, 5, 1)
$F_{1,3}$	11	(1, 3, 9, 8, 2, 6, 4, 4, 5, 8, 5)
$F_{2,3}$	7	(2, 6, 5, 3, 5, 4, 3)
$F_{3,4}$	13	(11, 1, 3, 4, 3, 1, 9, 9, 11, 11, 6, 1, 8)

Table 3: Complete root-free modular certificates for the second endpoints.

□

Corollary 11.3. *Every positive three-prime exponent triple with maximum minus minimum at most three gives a nonintegral ideal intersection graph.*

Proof. Repeated entries are covered by Theorem 7.1. After sorting distinct entries, the only possibilities have gaps $(d, e) = (1, 2), (1, 3), (2, 3)$, all covered by Theorem 11.2. □

11.1 The complete finite region for minimum exponents four through seven

The uniform cutoff makes the requested diagnostic range complete for each of the minimum exponents 4, 5, 6, 7. Here the relevant quintic is $g_B(x) = \det(xI - B)/x$ for H , whereas the preceding endpoint arguments use the complement quintic $h_C(x) = -g_B(N - x)$.

Corollary 11.4. *Every positive three-prime exponent triple with minimum entry at most seven gives a nonintegral ideal intersection graph.*

Proof. Theorem 10.2 handles minimum entries at most three. For a minimum entry $a \in \{4, 5, 6, 7\}$, repeated entries are covered by Theorem 7.1, and Proposition 10.1 settles $c \geq 4a^2 - 2a$. The remaining ordered triples lie in the exact finite domain $a < b < c < 4a^2 - 2a$.

For all 27 562 triples in this domain, the integer discriminant $\Delta_{a,b,c}$ of g_B is positive and strictly between two consecutive integer squares. Table 4 gives the complete counts. The certificate `results/open-region-discriminants.csv` records every triple, discriminant and integer q satisfying $q^2 < \Delta_{a,b,c} < (q+1)^2$. Each discriminant is independently recomputed as a 9×9 integer Sylvester determinant by `scripts/check_open_region_certificate.py`, using fraction-free elimination and exact divisions. The certificate exhausts the domain without omitted or duplicate triples.

If all roots of the monic quintic g_B were integers, its discriminant, the product of the squared pairwise root differences, would be an integer square. Thus every certified quintic has a noninteger root. The quotient is similar to a real symmetric Laplacian, so the root is real and nonzero; Lemma 2.1 gives a noninteger graph eigenvalue. This complete finite certificate and the written cutoff exhaust all triples with minimum entry at most seven. □

Minimum a	4	5	6	7	Total
Exclusive cutoff	56	90	132	182	—
Complete pairs	1275	3486	7750	15051	27562
Nonsquare discriminants	1275	3486	7750	15051	27562

Table 4: Exhaustive discriminant certificates in the region derived from the uniform cutoff.

The signs of $(g_B(1), g_B(2), g_B(3))$ are $(-, -, -)$ throughout this finite range. The complement signs instead have three patterns, $(-, +, -)$, $(-, +, +)$, $(+, +, +)$, recorded separately in the diagnostic. Different endpoint patterns do not rule out a uniform proof, and equal signs can miss two roots in one interval. Proposition 11.1 addresses that issue without extrapolating from these finite counts.

12 A uniform low eigenvalue and a balanced region

The Schur complement of Proposition 11.1 also counts roots when its diagonal has mixed signs. A comparison at three gives a low eigenvalue for every triple with minimum exponent at least four.

Lemma 12.1. *For $0 < x < a$, suppose all three $k_t(x)$ are nonzero. Let q be the number of negative diagonal values and put*

$$F(x) = 1 - \frac{a}{k_a(x)} - \frac{b}{k_b(x)} - \frac{c}{k_c(x)}.$$

Then $n_-(K(x)) = q + \mathbf{1}_{\{F(x) < 0\}}$ and nullity $K(x) = \mathbf{1}_{\{F(x) = 0\}}$. The number of positive complement quotient eigenvalues strictly below x is $n_-(K(x)) - 1$.

Proof. The bordered matrix

$$\begin{pmatrix} \text{diag}(k_t(x)) & v \\ v^T & 1 \end{pmatrix}$$

is congruent both to $K(x) \oplus [1]$ and to $\text{diag}(k_t(x)) \oplus [F(x)]$. Sylvester's law of inertia gives the formulas. The block $D - xI$ is positive definite, and C has a simple zero eigenvalue; subtracting this eigenvalue gives the stated count. $\square$

Theorem 12.2. *For every ordered triple $4 \leq a \leq b \leq c$, the six-support complement quotient has a nonzero eigenvalue in $(0, 3)$.*

Proof. Use $s = a + b + c$, $P = abc$ and $v = (\sqrt{a}, \sqrt{b}, \sqrt{c})^T$ from Proposition 11.1, and define

$$L = \text{diag} \left(s - \frac{3P}{a^2}, s - \frac{3P}{b^2}, s - \frac{3P}{c^2} \right) - vv^T.$$

Expansion of this diagonal-minus-rank-one determinant gives

$$\begin{aligned} \det L &= 6(c(a-b)^2 + b(a-c)^2 + a(b-c)^2), \\ v^T L v &= -3P \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) < 0. \end{aligned}$$

If the exponents are not all equal, the determinant is positive. The negative Rayleigh value supplies a negative eigenvalue, so the positive determinant forces exactly two negative eigenvalues. If all

exponents are equal, $L = -vv^T$, with one negative and two zero eigenvalues. In both cases there is a two-dimensional subspace on which L is nonpositive.

Since $a > 3$, the identity

$$K(3) = L - \text{diag}_{t \in \{a,b,c\}} \left(3 + \frac{9P}{t^2(t-3)} \right)$$

subtracts a positive definite matrix. Thus $K(3)$ is strictly negative on that subspace and has at least two negative eigenvalues. As $D - 3I$ is positive definite, $C - 3I$ also has at least two negative eigenvalues. The connected complement quotient has a simple zero eigenvalue and all other eigenvalues are positive. At least one of them therefore lies strictly between zero and three. This argument also covers zero values among the $k_t(3)$. $\square$

Corollary 12.3. *For $4 \leq a \leq b \leq c$, if*

$$(a-2)(a+b+c-2) > 2bc, \quad (12)$$

then C has an eigenvalue in $(2, 3)$, and $G_{p^a q^b r^c}$ is not Laplacian integral. Nonzero values of both $h_C(1)$ and $h_C(2)$ also imply nonintegrality. Hence $h_C(1)h_C(2) = 0$ is necessary for an integral graph.

Proof. Condition (12) is exactly $k_a(2) > 0$, so all three diagonal values are positive. As in Proposition 11.1, congruence turns $K(2)$ into $I - zz^T$, with $\|z\|^2 = \sum_t t/k_t(2) > 1$, since every $k_t(2) < s$. It has one negative and two positive eigenvalues and no zero eigenvalue. Consequently the only quotient eigenvalue at or below two is zero. The positive root below three from Theorem 12.2 lies in $(2, 3)$ and lifts to $(V-3, V-2)$. For the endpoint assertion, a positive root in $(0, 3)$ can be an integer only if it equals one or two, giving the necessary condition. $\square$

Corollary 12.4. *Let a be the minimum exponent and ρ the maximum minus minimum. Every positive three-prime exponent triple with $a \geq 3\rho + 8$ gives a nonintegral ideal intersection graph.*

Proof. Write the ordered triple as $(a, a+d, a+\rho)$ with $0 \leq d \leq \rho$. The left side minus the right side of (12) is

$$a^2 - 2a\rho - 8a - 2\rho^2 - 4\rho + 4 + (\rho-d)(a+2\rho+2).$$

The last term is nonnegative. At $a = 3\rho + 8 + u$, $u \geq 0$, the first part is $(\rho+2)^2 + (4\rho+8)u + u^2 > 0$. Corollary 12.3 applies. $\square$

For example, $(20, 21, 24)$, $(38, 43, 48)$ and $(308, 350, 408)$ satisfy the span criterion and lie outside the four fixed-gap families of Theorem 11.2. The argument allows an unbounded span and uses no exponent scan or modular endpoint certificate. The necessary endpoint-zero condition is not sufficient for integrality: those triples may still have other noninteger quotient eigenvalues.

13 Arithmetic obstructions from the complement quotient

The uniform low-root bound rules out common exponent divisors greater than two. Reduction of the characteristic polynomial gives further infinite congruence classes, even when another quotient root is an integer.

Theorem 13.1. *Every positive three-prime exponent triple with $\gcd(a, b, c) \geq 3$ gives a nonintegral ideal intersection graph.*

Proof. Put $d = \gcd(a, b, c)$. All entries of the six-support complement quotient C are integer combinations of the exponents and their pairwise products, so C/d is an integer matrix. Every rational eigenvalue of an integer matrix is an integer, by the rational-root theorem for its monic characteristic polynomial. Therefore every integer eigenvalue of C is a multiple of d .

Order $a \leq b \leq c$. If $a = 3$, Theorem 10.2 applies. Otherwise $a \geq 4$, and Theorem 12.2 supplies an eigenvalue $0 < \mu < 3$. No positive multiple of $d \geq 3$ lies in this interval. Hence μ and its graph lift $V - \mu$ are noninteger. $\square$

In particular, if m has exactly three distinct prime factors, then G_{m^d} is not Laplacian integral for every $d \geq 3$.

Theorem 13.2. *A positive three-prime exponent triple gives a nonintegral ideal intersection graph under either of the following conditions:*

1. *all three exponents are odd;*
2. *there is an odd prime ℓ and a nonzero residue t such that $a \equiv b \equiv c \equiv t \pmod{\ell}$ and $t^2 + 8t + 4$ is a quadratic nonresidue modulo ℓ .*

In particular, common residues 1, 3, 4 modulo 5, or 1, 2, 4, 5 modulo 7, imply nonintegrality.

Proof. The monic integer quintic $h_C(x) = \det(xI - C)/x$ would split into linear factors modulo every prime if all its roots were integers. A nonlinear irreducible factor in one reduction therefore supplies a noninteger real complement root and a noninteger graph lift.

If the exponents are odd, reduce the general quotient modulo two:

$$h_C(x) \equiv x(x^2 + x + 1)^2 \pmod{2}.$$

The quadratic has no root in $\mathbb{F}_2$, proving the first assertion. For a common residue t , the equal-exponent identity gives

$$h_C(x) \equiv (x - t^2 - t)(x^2 - (t^2 + 4t)x + 3t^2)^2 \pmod{\ell}.$$

The quadratic discriminant is $t^2(t^2 + 8t + 4)$. For odd ℓ and nonzero t , the assumed nonresidue makes the quadratic irreducible. Modulo five the discriminant factors for $t = 1, 3, 4$ are 3, 2, 2, respectively; the squares are 0, 1, 4. Modulo seven the corresponding values for $t = 1, 2, 4, 5$ are 6, 3, 3, 6, outside the square residues. $\square$

These conditions allow unequal exponents with arbitrarily large ratios or differences. For example, (11, 16, 21), (13, 18, 23) and (14, 19, 24) have gcd one and mixed parity, but meet the modulo-five criterion. The other parity reductions are x^5 for three even exponents and $x^2(x + 1)^3$ for one or two odd exponents. Their complete splitting does not establish integrality.

The arithmetic arguments do not assume that the smallest positive quotient root is noninteger. Scaling also yields an obstruction for even exponents whose binary reductions split completely.

Theorem 13.3. *Every positive three-prime exponent triple with $a \equiv b \equiv c \equiv 2 \pmod{4}$ gives a nonintegral ideal intersection graph.*

Proof. Write $(a, b, c) = (4u + 2, 4v + 2, 4w + 2)$, set $Q = C/2$, and put $f(x) = \det(xI - Q)/x$. The matrix Q is integral. If all eigenvalues of C were integers, those of Q would be rational and hence integers by the rational-root theorem.

Modulo two, Q is block lower triangular. Its upper block has zero diagonal and all off-diagonal entries one, and its lower block is I_3 . Consequently

$$\det(xI - Q) \equiv x(x+1)^5 \pmod{2}, \quad f(x) \equiv (x+1)^5 \pmod{2}.$$

Thus all five hypothetical integer roots of f are odd. For every odd integer t , their five factors $t - \lambda_i$ would give $32 \mid f(t)$. On the other hand, expansion of the general scaled quotient gives the coefficientwise polynomial congruence

$$f(2y+1) \equiv 8(u+v+w+y-1) \pmod{32}.$$

In particular $f(3) - f(1) \equiv 8 \pmod{32}$, contradicting the required divisibility of both values. A noninteger complement eigenvalue and its noninteger graph lift follow. $\square$

Corollary 13.4. *If the three positive exponents have the same 2-adic valuation, their ideal intersection graph is not Laplacian integral.*

Proof. At valuation zero, Theorem 13.2 applies. At valuation one, Theorem 13.3 applies. At valuation at least two, their common divisor is at least four, so Theorem 13.1 applies. $\square$

For example, $(10, 14, 18)$, $(14, 22, 34)$ and $(18, 26, 42)$ have gcd two and lie below the linear even-tail cutoff of the following section. The new obstruction applies regardless of the minimum exponent or ratios. The necessary endpoint-zero condition of Corollary 12.3 remains insufficient: the already settled boundary triple $(9, 9, 136)$ has $h_C(1) = 0$ and other noninteger quotient roots. The remaining three-prime problem has gcd at most two, at least one even exponent, and lies outside the congruence classes above as well as the earlier proved regions.

13.1 A mixed-parity divisibility obstruction

The distribution of hypothetical integer roots modulo four gives further infinite classes with unequal exponent valuations.

Lemma 13.5. *Let $f \in \mathbb{Z}[x]$ be monic of degree five with $f(x) \equiv x^2(x+1)^3 \pmod{2}$. If all its roots are integers, then $32 \mid f(1)$ or $32 \mid f(3)$.*

Proof. The binary factorization forces precisely three odd roots and two even roots, counted with multiplicity. At least two odd roots share a residue $r \in \{1, 3\}$ modulo four. In $f(r)$, their two factors are divisible by four and the third odd-root factor is divisible by two. The even-root factors are odd. Therefore $32 \mid f(r)$, including when $f(r) = 0$. $\square$

Theorem 13.6. *Let a, b be the two odd exponents and c the even exponent of a positive three-prime exponent triple. Each row in the following table, including exchange of a, b , implies that its ideal intersection graph is nonintegral.*

$a \bmod 8$	$b \bmod 8$	$c \bmod 8$	$h_C(1) \bmod 32$	$h_C(3) \bmod 32$
1	3	2	8	16
1	7	2	24	16
3	7	2	16	8
3	5	6	8	16
3	7	6	16	24
5	7	6	24	16

Proof. For either mixed exponent parity, $h_C(x) \equiv x^2(x+1)^3 \pmod{2}$. Substituting $(a, b, c) = (8u+r_1, 8v+r_2, 8w+r_3)$ in the general quotient quintic gives the last two columns: every nonconstant coefficient in u, v, w is divisible by 32. Both entries are nonzero modulo 32, contradicting Lemma 13.5 if all quotient roots were integers. The resulting noninteger real complement eigenvalue has a noninteger graph lift. $\square$

For example, $(10, 17, 19)$ and $(14, 19, 21)$ meet these criteria with gcd one and unequal 2-adic valuations. There is no minimum or ratio bound in the argument. The weaker divisibilities $4 \mid h_C(2t)$ and $8 \mid h_C(2t+1)$ hold identically for both mixed exponent parities; they alone do not supply the contradiction. The distribution of the three odd roots modulo four is the additional constraint.

Lemma 13.7. *Let $f \in \mathbb{Z}[x]$ be monic of degree five with $f(x) \equiv x^2(x+1)^3 \pmod{2}$. Suppose $F(y) = f(2y+1)/8$ belongs to $\mathbb{Z}[y]$ and $F(y) \equiv y^3 \pmod{2}$. If all roots of f are integers, then its three odd roots are congruent to one modulo four, $64 \mid f(1)$ and $16 \mid f'(1)$.*

Proof. Write the two even roots as η_1, η_2 and the three odd roots as $2r_1+1, 2r_2+1, 2r_3+1$, with multiplicities. Then

$$F(y) = \prod_{j=1}^2 (2y+1-\eta_j) \prod_{i=1}^3 (y-r_i).$$

The first two factors reduce to one modulo two. Hence $\prod_i (y-r_i) \equiv y^3 \pmod{2}$, which forces every r_i to be even. Each odd-root factor in $f(1)$ is therefore divisible by four, giving $64 \mid f(1)$. Every term in the product-rule expansion of $f'(1)$ retains at least two such factors, giving $16 \mid f'(1)$. The argument includes coincident roots and zero factors. $\square$

Theorem 13.8. *Every positive three-prime exponent triple that is a permutation of $(3, 3, 2)$ modulo four has a nonintegral ideal intersection graph.*

Proof. Relabel the odd exponents as $a = 4u+3$, $b = 4v+3$ and the even exponent as $c = 4w+2$, where $u, v, w \geq 0$. Expansion of the general complement quintic gives the coefficientwise identities

$$\begin{aligned} h_C(2y+1)/8 &\in \mathbb{Z}[u, v, w, y], & h_C(2y+1)/8 &\equiv y^3 \pmod{2}, \\ h_C(1) &\equiv 16(u+v) \pmod{32}, & h'_C(1) &\equiv 8(u+v+1) \pmod{16}. \end{aligned}$$

Its binary reduction is $x^2(x+1)^3$. If all five roots were integers, Lemma 13.7 would give $64 \mid h_C(1)$, so the endpoint congruence would require $u+v$ even. The derivative divisibility $16 \mid h'_C(1)$ would require $u+v$ odd, a contradiction. Thus there is a noninteger real complement eigenvalue, whose graph lift is noninteger. $\square$

This theorem contains the two rows $(3, 7, 2)$ and $(3, 7, 6)$ of Theorem 13.6 and also covers odd exponents sharing the same residue modulo eight. For example, $(10, 19, 27)$ and $(14, 23, 39)$ have odd residues $(3, 3)$ and $(7, 7)$ modulo eight, respectively. Neither theorem requires a low-root interval or an exponent-size bound.

For the fixed necessary divisor 32, increasing the exponent modulus alone gives no further exclusions. On all mixed-parity triples, the values of both $h_C(1)$ and $h_C(3)$ modulo 32 already have period eight in each exponent. This follows coefficientwise by substituting the two parity representatives $(2u+1, 2v, 2w)$ and $(2u+1, 2v+1, 2w)$ and shifting one variable by $4z$; symmetry covers their permutations. Thus the higher-modulus tables are lifts of the six classes in Theorem 13.6. The additional exclusions in Theorem 13.8 use the distribution of the roots and the derivative, beyond those two endpoint divisibilities.

14 A linear bound for the endpoint two

The two possible low integer roots have different arithmetic constraints. The endpoint two is confined to a linear region in the minimum exponent; this gives a new nonintegral tail for even exponent triples.

Theorem 14.1. *Let $4 \leq a \leq b \leq c$, and put*

$$R(a) = \frac{7a^2 - 2a + 8}{a + 2} = 7a - 16 + \frac{40}{a + 2}.$$

If $h_C(2) = 0$, then $c < R(a)$. In fact, $c \geq R(a)$ implies $h_C(2) > 0$.

Proof. Fix a, c and expand the symmetric endpoint polynomial in b :

$$h_C(2) = A(a, c)b^3 + B(a, c)b^2 + D(a, c)b + E(a, c),$$

where

$$\begin{aligned} A(a, c) &= ac(a + c - 1) + 2a(a - 1) + 2c(c - 1) - 4, \\ B(a, c) &= c^2((a + 2)c - (7a^2 - 2a + 8)) \\ &\quad + (a^3 + 2a^2 - 6a - 4)c + 2(a - 2)(a^2 - 2a - 6), \\ D(a, c) &= (a - 2)(c - 2)(a^2c + a^2 + ac^2 + 6ac + 4a + c^2 + 4c - 12), \\ E(a, c) &= 2(a - 2)(c - 2)(a + c - 2)(ac + a + c - 2). \end{aligned}$$

For $a, c \geq 4$, the coefficients A, D, E are strictly positive. Both $a^3 + 2a^2 - 6a - 4$ and $a^2 - 2a - 6$ are positive at $a = 4$ and increasing on $[4, \infty)$. Thus $c \geq R(a)$ also gives $B > 0$, which proves the strict endpoint sign. Conversely, an endpoint zero requires $B < 0$, so $(a + 2)c < 7a^2 - 2a + 8$. $\square$

Corollary 14.2. *If $4 \leq a \leq b \leq c$ are all even and $c \geq R(a)$, then $G_{p^a q^b r^c}$ is not Laplacian integral. For minimum exponent at least eight, $c \geq 7a - 12$ suffices.*

Proof. Every entry of C is even, so every integer eigenvalue of C is even by the rational-root theorem applied to $C/2$. In particular, one is not an eigenvalue. Theorem 14.1 excludes two, and Theorem 12.2 provides a noninteger root in $(0, 3)$. Equivalently, $h_C(0) < 0 < h_C(2)$ directly yields a root in $(0, 2)$, which cannot equal one. Its graph lift is noninteger. For $a \geq 8$, $40/(a + 2) \leq 4$ gives $R(a) \leq 7a - 12$. $\square$

The triples $(8, 10, 44)$, $(10, 14, 58)$ and $(40, 42, 266)$ meet this criterion with gcd two. Their largest exponents are below the earlier quadratic cutoff and their spans are outside the earlier balanced region. The already settled repeated triple $(10, 10, 12)$ has $h_C(2) = 0$ and obeys the strict bound; an integer endpoint does not establish integrality.

Proposition 14.3. *For fixed $a, b \geq 4$, any positive integer c on an endpoint-zero surface satisfies*

$$\begin{aligned} h_C(1) = 0 &\implies c \mid (a - 1)(b - 1)(a + b - 1)(ab + a + b - 1), \\ h_C(2) = 0 &\implies c \mid 2(a - 2)(b - 2)(a + b - 2)(ab + a + b - 2). \end{aligned}$$

Proof. Both endpoints are integer cubics in c . Their constant terms are the displayed nonzero products. Reduce each endpoint equation modulo c . $\square$

For a fixed pair this gives finite divisor candidates, which must still satisfy the endpoint equation. Reza's requested diagnostic for middle exponents through fifteen now has a complete independent certificate.

Corollary 14.4. *Every positive ordered three-prime exponent triple $a \leq b \leq c$ with $b \leq 15$ gives a nonintegral ideal intersection graph. This conclusion uses the complete finite endpoint certificate below.*

Proof. Repeated exponents are covered by Theorem 7.1, and minimum exponents one, two and three by Theorems 8.1, 9.1 and 10.2, respectively. It remains to consider $4 \leq a < b \leq 15$ and $c > b$.

For each of these 66 pairs, both endpoint polynomials are integer cubics in c with the positive constant terms in Proposition 14.3. The complete endpoint certificate records all 132 coefficient lists and exhausts their 8 527 positive divisors using integer trial division. Independent integer Horner evaluation at every one of the 7 585 divisors above b gives no zero. Thus neither endpoint vanishes for any integer $c > b$, with no imposed upper bound on c . Theorem 12.2 supplies a positive root in $(0, 3)$; it can equal neither one nor two, proving nonintegrality. $\square$

The reflected quotient in this diagnostic uses $N = |V(H)| = a + b + c + ab + ac + bc$, not the full graph vertex count. The inclusive requested bound $b \leq 15$ and exact rational linear-factor extraction are cross-checked against the independent divisor certificate. The empty lists settle this pair range; they do not prove either endpoint surface globally empty.

Theorem 14.5. *Every positive three-prime exponent triple congruent to $(2, 2, 2)$, or to a permutation of $(1, 2, 2)$, modulo three gives a nonintegral ideal intersection graph.*

Proof. For these two residue patterns, reduction of the general complement quintic gives, respectively,

$$h_C(x) \equiv x^5, \quad h_C(x) \equiv (x^2 + 1)(x^3 + x^2 - x + 1) \pmod{3}.$$

The corresponding pairs $(h_C(1), h_C(2))$ are $(1, 2)$ and $(1, 1)$ modulo three, so neither integer endpoint is a root. Minimum exponents at most three are already covered. Otherwise Theorem 12.2 supplies a positive root in $(0, 3)$, which cannot be an integer. Its graph lift is noninteger as well. For the second pattern, the irreducible factor $x^2 + 1$ also directly rules out a completely integral quotient spectrum. $\square$

For example, $(8, 17, 22)$ and $(8, 17, 23)$ satisfy the new conditions with gcd one and unequal 2-adic valuations. Complete endpoint tables modulo two and three, checked against independent integer determinants, are given in the residue certificate. Every pair (a, b) modulo either prime has at least one c root on each surface; retaining the full root sets, rather than only root-free pairs, is necessary to detect these exclusions.

Remark 14.6. Endpoint tests at a fixed finite set of moduli cannot by themselves prove either surface empty on all fully distinct positive triples. Let M be their least common multiple. Since $(9, 9, 136)$ has $h_C(1) = 0$, the strictly ordered triples $(9 + kM, 9 + (k + 1)M, 136 + (k + 2)M)$ pass every selected endpoint-one test. The known endpoint-two zero $(10, 10, 12)$ similarly gives survivors $(10 + kM, 10 + (k + 1)M, 12 + (k + 2)M)$. For large k their middle exponents exceed fifteen. These are congruence survivors, not asserted integer zeros. Congruences can still be combined with the earlier interval restrictions or an obstruction from another quotient root.

The remaining all-even triples have gcd two, $8 \leq a < b < c < R(a)$ and $b \geq 16$, outside the preceding proved regions. The mixed-parity surface $h_C(1) = 0$ still needs another obstruction.

15 Complete fixed-exponent families and exact diagnostics

Theorem 15.1. *For every $a \in \{2, 3, 4\}$, every $b \geq 1$ and distinct primes p, q, r , the graph $G_{p^a q^a r^b}$ is not Laplacian integral.*

Proof. The cutoff values are $R(2) = 0$, $R(3) = 4/5$, and $R(4) = 2$. Theorem 6.4 settles all b for $a = 2, 3$, and all $b > 2$ for $a = 4$. At $(a, b) = (4, 1)$, $D = 145$ is nonsquare. At $(4, 2)$ the symmetric cubic is

$$f_{4,2}(x) = x^3 - 86x^2 + 2304x - 18816.$$

Its values at $0, 1, 2, 3, 4$ modulo 5 are $4, 3, 1, 4, 3$, so it has no root modulo 5. A cubic over a field without a root is irreducible. Since this cubic is monic, reduction modulo 5 proves irreducibility over $\mathbb{Q}$. Its real nonzero roots lift to noninteger graph eigenvalues. $\square$

Corollary 15.2. *The same all- b conclusion holds for $a = 5, 6$. Thus it holds for every $a \in \{2, 3, 4, 5, 6\}$.*

Proof. The second cutoff gives $S(5) = 14/13$ and $S(6) = 9/5$, leaving only $b = 1$ in each case. The discriminants are 221 and 313, respectively. Since $14^2 < 221 < 15^2$ and $17^2 < 313 < 18^2$, both give irrational antisymmetric eigenvalues. Every larger b is covered by Theorem 6.5. $\square$

Corollary 15.3. *Every positive b also gives a nonintegral spectrum for $a = 7, 8, 9$. Thus every repeated exponent $a \in \{2, 3, 4, 5, 6, 7, 8, 9\}$ is settled.*

Proof. The third cutoff gives $T(7) = 6/5$, $T(8) = 12/7$, $T(9) = 70/31$. Only four pairs remain under these bounds:

$$(a, b) = (7, 1), (8, 1), (9, 1), (9, 2).$$

Their discriminants are 421, 545, 685, 1489, respectively, between the consecutive squares with lower square roots 20, 23, 26, 38. Each therefore gives irrational antisymmetric eigenvalues, and Theorem 6.7 covers every larger b . $\square$

For comparison, the complete divisor certificates before applying the linear cutoff are given in Table 5. The constants factor as $M = 40 = 2^3 \cdot 5$, $189 = 3^3 \cdot 7$, and $576 = 2^6 \cdot 3^2$. Testing every divisor $d \leq \sqrt{M}$ in (5) gives exactly the displayed admissible pairs. The corresponding cubics in Table 6 have no root modulo the listed primes. These finite certificates supply a second proof using the complete exceptional list. Three pairs are in the boundary family; $(4, 2)$ is the additional symmetric-block obstruction.

a	M	(d, e)	b	$\sqrt{D}$
2	40	(1, 40)	3	13
3	189	(1, 189)	10	47
4	576	(6, 96)	2	18
4	576	(1, 576)	21	115

Table 5: Every admissible factor pair for $a = 2, 3, 4$.

The separate diagnostic range $a = 2, 3, 4$, $1 \leq b \leq 30$ contains 90 cubics, each with a noninteger root certified in an interval between consecutive integers. The reducible cases in that range are $b = 2, 6$ for $a = 2$, $b = 3, 12$ for $a = 3$, and $b = 4, 20$ for $a = 4$. Each has a linear and an irreducible

(a, b)	$f_{a,b}(x)$	Witness prime
$(2, 3)$	$x^3 - 47x^2 + 702x - 3312$	5
$(3, 10)$	$x^3 - 187x^2 + 11220x - 214200$	13
$(4, 2)$	$x^3 - 86x^2 + 2304x - 18816$	5
$(4, 21)$	$x^3 - 485x^2 + 75492x - 3721536$	5

Table 6: Root-free modular reductions for the exceptional cubics.

quadratic factor. These observations do not require whole-cubic irreducibility. Indeed, for every $a \geq 1$,

$$f_{a,a}(x) = (x - 2a(a + 1))(x^2 - (5a^2 + 2a)x + 6a^4 + 3a^3),$$

$$f_{a,a(a+1)}(x) = (x - 2a(a + 1)^2)(x^2 - (3a^3 + 4a^2 + 2a)x + 2a^6 + 6a^5 + 7a^4 + 3a^3).$$

Consequently no prime can make every diagnostic cubic irreducible: the monic factors in the diagonal family retain positive degree modulo every prime.

For the second family the quadratic discriminant is $a^2(a^4 + 4a + 4)$. For $a \geq 3$ the radicand is between a^4 and $(a^2 + 1)^2$, since $2a^2 - 4a - 3 = 2(a - 3)^2 + 8(a - 3) + 3 > 0$. At $a = 2$ it is 28, between 5^2 and 6^2 . Thus the quadratic is irreducible for every $a \geq 2$.

Remark 15.4. At $a = 1, b = 2$ the cubic is $(x - 3)(x - 6)(x - 8)$, although the full graph has irrational antisymmetric eigenvalues $(13 \pm \sqrt{33})/2$. A direct construction of its 10 vertices gives

$$\det(xI - L) = x(x - 10)(x - 9)^3(x - 7)^2(x - 4)(x^2 - 13x + 34).$$

Thus a noninteger cubic root is a sufficient condition for graph nonintegrality; the converse cannot be used without the other blocks.

16 Exact verification and reproducibility

The infinite-family conclusions follow from the proofs above, including the explicitly exhausted finite certificates in the repeated-exponent and minimum-three and minimum-seven arguments. The accompanying computations use integer or rational arithmetic to verify embeddings, identities and finite certificates. In particular, the all- b result for $a = 2, 3, 4$ uses the exhaustive divisor criterion, rather than extrapolation from the 90 diagnostic cases. No Lean formalization is claimed for these Laplacian results.

The code and saved JSON certificates are available in the public project repository. The build instructions beside this manuscript give commands and record the exact source revisions. The checks have the following scopes.

1. `scripts/check_integrality_obstructions.py` constructs ideals as exponent tuples and tests adjacency by coordinatewise maxima, independently of the support description. It verifies $LU = UB$ for the displayed invariant embeddings, for squarefree $t = 3, \dots, 8$ and $(1, 1, k)$ with $k = 1, \dots, 8$: 14 cases and 44 294 vertex pairs. It also checks the square-gap inequalities and principal-minor witnesses.
2. `scripts/check_remaining_case.py` constructs all 34 vertices for $(2, 2, 3)$ and verifies its full characteristic polynomial and the class decomposition in exact arithmetic. The shifted symmetric cubic is $x^3 - 80x^2 + 2099x - 18052$; its reduction modulo 5 has no root.

3. `scripts/check_repeated_exponent_family.py` checks the general quotient polynomial, both boundary sign identities, the positive-coefficient expansion, and the integral antisymmetric eigenvalues symbolically. Direct graphs for $(2, 2, 3)$, $(3, 3, 10)$, $(4, 4, 21)$ have 34, 174, 548 vertices, covering 165 490 vertex pairs. Exact neighbor counts verify both blocks and their lifting. The check computes the seven-support quotient polynomial; it does not compute the full 548×548 characteristic polynomial.
4. `scripts/check_aa_b.py` factors the 90 diagnostic cubics over $\mathbb{Q}$ and uses exact rational root isolation for a root between consecutive integers. Independent Sturm counts and endpoint signs were checked for every saved interval. Its modular search is restricted to 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31; absence of a listed witness is not a proof that no prime works for that cubic.
5. `scripts/check_fixed_repeated_exponents.py` reconstructs the cubic from B , checks the discriminant, product, cutoff and reducible factor identities, and enumerates every factor pair for $a = 2, 3, 4$. It verifies the admissibility conditions and all modular witnesses in Tables 5 and 6. The general congruence and factor inequality are proved in Section 6.
6. `scripts/check_second_cutoff.py` checks the polynomial remainder modulo $a + 2$, enumerates every divisor of 24, and verifies the four root-free cubic reductions settling all $d = a + 2$ cases. It also checks the second-cutoff identities and the two nonsquare discriminants completing $a = 5, 6$. This finite certificate follows the proved divisor reduction; it is not a parameter-range search.
7. `scripts/check_factor_index_reduction.py` verifies the general fixed-index divisibility and reconstruction identities, exhausts the divisors of $K_2 = 108$, and certifies the unique positive- b index-2 cubic modulo 13. It checks the third-cutoff identities and the four remaining discriminants for $a = 7, 8, 9$. These finite checks exhaust the proved candidate sets, with no arbitrary parameter cutoff.
8. `scripts/check_repeated_exponent_completion.py` reconstructs the general cubic from the 4×4 quotient, checks both sign-interval identities, and verifies the fourth-cutoff and strictly positive gap identities. It exhausts every divisor of 24, 108, 320 for indices 1, 2, 3, reproducing all eight positive- b modular cubic certificates with separate integer Horner arithmetic. The infinite remaining range is settled by the sign proof of Theorem 7.1, not by a scan.
9. `scripts/check_six_support_quotient.py` verifies the general six-support and complement quotients, their weighted symmetry and reflected quintics, and the two determinant values used in Theorem 8.1. Exact rational factorization covers all thirty-five requested triples $1 \leq a < b < c \leq 7$, with every matrix and quintic cross-checked against `scripts/check_fully_distinct.py`. All saved rational intervals pass independent Sturm counts and endpoint-sign checks. Direct ideal adjacency verifies the quotient for $(1, 2, 3)$ and $(2, 3, 4)$, using 22 and 58 vertices and 312 representative-to-vertex pairs. The infinite unit-exponent conclusion uses the written endpoint proof.
10. `scripts/check_minimum_two.py` verifies the complement endpoint identities at $a = 2$, the symmetric expression in $b + c$ and bc , and both positive-coefficient expansions that exhaust the infinite range. It reconstructs the single exceptional quintic at $(2, 3, 4)$ and checks its rational endpoint values with separate Horner evaluation and a Sturm count. No new parameter range is enumerated.

11. `scripts/check_distinct_tail.py` reconstructs the general six-support complement quintic and checks its cubic first endpoint, the uniform cutoff identity and positive remainder, and both minimum-three second-endpoint expansions. The cutoff itself derives the finite domain $4 \leq b < c < 30$, containing exactly 325 pairs. For every pair the checker reconstructs the integer quotient quintic, matches its symbolic specialization, and checks the displayed root interval using separate rational Horner arithmetic. Separate cubic Horner evaluation confirms all first-endpoint values: 257 positive, 68 negative and no zero. The $(3, 4, 5)$ rational exception also has a Sturm count of one in $(1, 3/2)$. The minimum-three theorem uses this complete finite certificate; the general cutoff uses the written positive-coefficient argument.
12. `scripts/check_low_spectrum.py` reconstructs the weighted symmetric complement quotient and checks its block form, Schur complement, determinant identity and ordered-parameter derivative. For the four bounded-gap families it verifies the upper-inertia expressions and both endpoint factorizations, positive shifted first-endpoint coefficients, and all 38 root-free modular residues with separate integer Horner arithmetic. Eight chosen representatives, the four families at $a = 4, 20$, have independently reconstructed integer quotient quintics and Sturm counts of two roots in $(0, 3)$; at $a = 20$ both lie in $(2, 3)$. These examples check the formulas. The infinite families use the written inertia proof and the complete modular exclusions, with no parameter scan.
13. `scripts/check_open_region2.py` runs the coauthor-requested domain $4 \leq a \leq 7, a < b < c < 4a^2 - 2a$. It constructs the generic H quotient determinant once and specializes its integer coefficients for every one of the 27 562 triples. Every discriminant is nonsquare. Original-quotient and complement endpoint signs are reported separately; the general reflection identity is checked symbolically. Sixteen representatives, including every first complement sign-pattern witness, have separately reconstructed integer matrix characteristic polynomials, complement endpoints and exact rational factorizations. The compatibility entrypoint `scripts/check_open_region.py` runs the same corrected code.
14. `scripts/check_open_region_certificate.py` independently reconstructs every one of the 27 562 discriminants as a 9×9 integer Sylvester determinant, using a separate fraction-free Bareiss implementation. It checks every exact division, every strict consecutive-square bracket and the complete ordered domain. This is the exhaustive finite certificate used in Corollary 11.4, combined with the written cutoff, rather than extrapolation to arbitrary minimum exponent.
15. `scripts/check_mixed_inertia.py` verifies the comparison determinant, negative Rayleigh value, equal-exponent case, positive comparison gap and balanced-span identities symbolically in a rational 3×3 congruence. Six exact secular/Sturm fixtures cover every nonsingular diagonal-sign/secular-sign combination possible here. Nine separately reconstructed quotient/Sturm fixtures include equal exponents, two zero-diagonal boundaries, a strongly unbalanced triple and chosen spans up to 1000. The finite fixtures validate formulas; Theorem 12.2 and Corollaries 12.3–12.4 use the written comparison and positivity proofs, with no parameter scan.
16. `scripts/check_arithmetic_obstructions.py` verifies symbolic integer quotient scaling, the equal-exponent quintic factorization and its quadratic discriminant. It checks all eight parity patterns and every nonzero common residue for primes 3, 5, 7, with all 70 quadratic residue evaluations, including the binary factor, cross-checked by integer Horner arithmetic. Six direct quotient fixtures verify scaling, low-root and modular witnesses, and the existing boundary triple $(9, 9, 136)$ checks that an integer endpoint is compatible with other noninteger roots.

The infinite conclusions use the written divisibility and finite-field proofs, with no exponent scan.

17. `scripts/check_endpoint_reduction.py` reconstructs the general endpoint-two cubic in the middle exponent and the two endpoint constant terms in the largest exponent. It verifies both rational-cutoff identities and five positive shifted-coefficient expressions. Three chosen even gcd-two fixtures below the earlier quadratic cutoff have direct integer quotient, independent Horner and exact Sturm checks. The existing repeated triples $(9, 9, 136)$ and $(10, 10, 12)$ check the endpoint divisibility conditions and retain nonlinear rational factors; the latter checks the strict linear bound. The infinite bound and even-tail corollary use written positivity and divisibility proofs, without an exponent scan or divisor-set enumeration.
18. `scripts/check_even_exponent_congruence.py` verifies integer quotient scaling, the independent binary block characteristic polynomial, and the coefficientwise congruences modulo two and thirty-two for the scaled quintic. Three chosen gcd-two triples inside the preceding even linear bound have separately reconstructed integer quotients, rational factorizations and independent integer Horner values at one and three. The infinite congruence-family theorem follows from the written divisibility contradiction, without an exponent scan.
19. `scripts/check_endpoint_surfaces.py` is Reza's requested diagnostic, corrected to use $|V(H)|$ in the complement reflection and to include $b = 15$. Exact rational linear-factor extraction finds no endpoint roots for $4 \leq a < b \leq 15$, $c > b$. `scripts/verify_endpoint_surfaces.py` independently reconstructs the direct disjoint-support complement matrix and all 132 endpoint cubics over the 66 pairs. It exhausts 8 527 positive constant-term divisors by integer trial division and checks all 7 585 divisors above b with separate integer Horner arithmetic. Two previously settled repeated-exponent positive controls recover their endpoint roots and match full quotient factorizations. This complete finite certificate, combined with the uniform low-root theorem and earlier cases, proves Corollary 14.4. There is no upper cutoff on c or extension beyond the requested pair range.
20. `scripts/check_endpoint_residues.py` reconstructs the general endpoint cubics and records all 26 pair/endpoint rows for primes two and three, including zero residues and zero polynomials. At all 35 positive residue representatives, a separate 720-term integer determinant expansion checks both endpoint values, for 70 checks. The eight parity reductions and five simultaneous modulo-three exclusions are verified against direct quotient characteristic polynomials. Two selected fully distinct triples have independent endpoint determinants and exact Sturm counts; two known repeated-exponent controls retain their endpoint zeros. The infinite classes in Theorem 14.5 use the written low-root proof. This is a residue enumeration, not an exponent search.
21. `scripts/check_mixed_parity_congruence.py` checks the two mixed binary reductions and four initial coefficient-divisibility identities. It verifies all 558 coefficients in twelve substitutions for the six rows of Theorem 13.6. Six positive residue representatives and two selected fully distinct triples have 16 independent 720-term integer determinant checks at one and three, matching separate integer Horner values. The fixtures retain nonlinear rational factors. Complete lower-modulus tables give compatible linear root multisets for all 12 and 80 unordered mixed-parity exponent classes modulo four and eight, respectively; this compatibility is not an integrality claim. The infinite theorem follows from the written three-odd-root divisibility lemma.

22. `scripts/check_odd_root_distribution.py` verifies the generic quintic's six exponent permutations and the coefficientwise identities in Theorem 13.8: 160 coefficients for the shifted polynomial's divisibility by eight, 160 for its reduction modulo two, 42 endpoint coefficients modulo thirty-two and 51 derivative coefficients modulo sixteen. Four selected fixtures have 28 independently expanded integer six-by-six determinants, including six interpolation points and the zero-root check per fixture. A further 24 five-by-five principal cofactors independently reconstruct the derivative at one. The reconstructed quintics and derivatives agree; the infinite theorem uses the written odd-root distribution argument.
23. Reza's `scripts/check_2adic_higher_moduli.py` was run unchanged at exponent moduli eight, sixteen and thirty-two. The independent `scripts/verify_2adic_higher_moduli.py` checks all 28,032 ordered mixed-parity residue triples using 56,064 integer Bareiss determinants for the two endpoints. Twelve coefficientwise identities, with 560 coefficients, establish coordinate period eight modulo thirty-two. The complete 384 base rows therefore encode the higher tables. The endpoint-divisibility test excludes 36, 288 and 2304 classes, respectively. These are finite residue tables, and their survivors do not assert integer endpoint zeros or integral spectra.

The symbolic and characteristic-polynomial checks use SymPy 1.14.0; the first checker uses the Python standard library. Saved results contain source hashes. The 10-vertex polynomial in the preceding remark was independently checked by a direct ideal-graph construction; it is also reproducible by the command in the manuscript build guide. No floating-point eigenvalue calculation is used as integrality evidence.

17 Remaining questions

Theorem 7.1 settles every triple with at least two equal exponents, and Corollary 11.4 settles every triple with minimum exponent at most seven. Proposition 10.1 settles every ordered triple with $c \geq 4a^2 - 2a$. Theorem 11.2 settles four further unequal-exponent families, including every triple with maximum minus minimum at most three. The remaining three-prime cases therefore lie within $8 \leq a < b < c < 4a^2 - 2a$, outside those families and the triples meeting Proposition 11.1 or Corollary 12.3. In particular, they satisfy $b \geq 16$ by the complete endpoint certificate of Corollary 14.4, as well as $(a - 2)(a + b + c - 2) \leq 2bc$ and $a < 3(c - a) + 8$. These cases and nonsquarefree exponent vectors with more than three prime factors require further arguments. The results here therefore do not establish nonintegrality for every integer with at least three distinct prime factors. Theorems 13.1 and 13.2 additionally restrict the remaining three-prime triples to gcd one or two, at least one even exponent, and residues outside the proved congruence classes. Theorem 14.5 also excludes the residue patterns $(2, 2, 2)$ and all permutations of $(1, 2, 2)$ modulo three. Corollary 13.4 also requires unequal 2-adic valuations. In the remaining all-even region at least one exponent is congruent to two modulo four and at least one is divisible by four. Theorem 13.6 also excludes six mixed-parity residue classes modulo eight, without a size bound. Theorem 13.8 excludes every permutation of $(3, 3, 2)$ modulo four, including cases with the two odd exponents equal modulo eight.

The remaining three-prime question is whether $G_{p^a q^b r^c}$ is always nonintegral in this remaining region. For each fixed minimum exponent a , the cutoff leaves a finite set of pairs; it does not bound a itself. The coordinate-swap decomposition used for repeated exponents is unavailable there. The full six-support quotient and its diagonal-minus-rank-one Schur complement provide exact tools for the remaining argument. Theorem 12.2 now places a positive quotient eigenvalue in $(0, 3)$

throughout the remaining region. A possible integral graph must therefore satisfy $h_C(1)h_C(2) = 0$. Theorem 14.1 confines the second surface to $c < 7a - 16 + 40/(a + 2)$. For all-even triples one is excluded, so the remaining gcd-two even region has this linear upper bound as well. Proposition 14.3 gives finite divisor candidates for either surface at a fixed pair (a, b) . Excluding integral triples on these two endpoint-zero surfaces subject to the remaining arithmetic restrictions, or finding another noninteger quotient root there, is the next focused question. The necessary endpoint-zero condition is not itself an integrality characterization.

The distinction between the two repeated-exponent blocks remains essential: an integral anti-symmetric block occurs on the boundary family, while an integral symmetric cubic occurs at $(1, 1, 2)$. The completion theorem uses both blocks, rather than asserting that either alone always obstructs integrality.

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